

4 Animal species have different resting heart rates.

- (a) The table shows the resting heart rate and mean life expectancy for some animal species.

Animal species	Resting heart rate / bpm	Mean life expectancy / years	Mean mass / kg	Mean resting metabolic rate / $\text{dm}^3 \text{O}_2 \text{kg}^{-1} \text{hr}^{-1}$
elephant	30	80	5780	1400
horse	38	30	500	910
human	72	78	110	360
rabbit	205	12	7	48
rat	420	1.8	2	10.8
humming bird	1260	7	0.02	0.012

- (i) Describe the relationships between resting heart rate and the data shown in the table.

(2)

.....

.....

.....

.....

.....

.....

- (ii) Calculate the volume of oxygen produced each hour by a resting elephant with a mass of 4500 kg.

Give your answer in standard form.

(3)

Answer

(b) The photograph shows a cheetah.



(Source: © Stu Porter/Shutterstock)

The heart rate for a cheetah increases from 120 bpm to 250 bpm during a chase. It can reach speeds of 100 km hr⁻¹

(i) Explain how the heart rate of a cheetah can be increased during a chase.

(4)

This image shows a full page of white paper with ten horizontal dashed lines, evenly spaced from top to bottom. The lines are thin and black, typical of handwriting practice paper. There is no text or other markings on the page.

(ii) Suggest why a cheetah can maintain this heart rate and speed for only a short period of time.

(2)

(Total for Question 4 = 11 marks)
